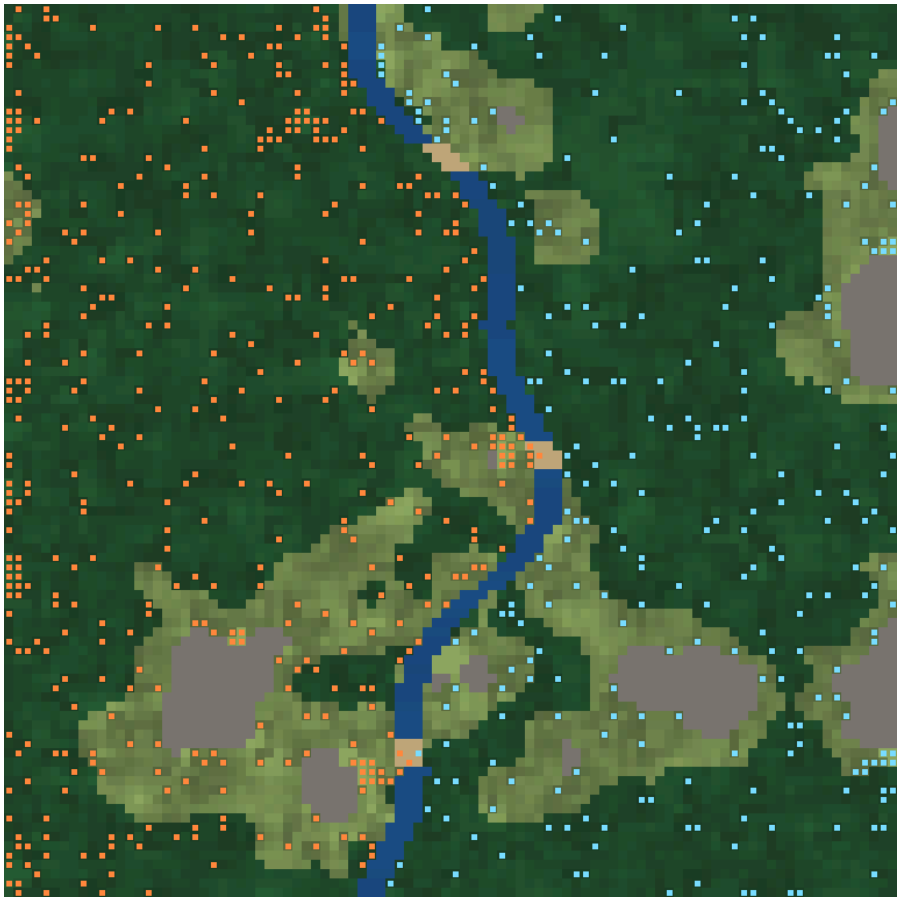
SETUP PAPER · SCENARIOS A + B · PRINT

River World on the Desktop

Scenarios A and B in one sealed world, with a river down the middle

11 OCTOBER 2026



COVERS

Install · four first-run checks · sixteen seeds on a Ryzen 9 5950X · the first experiment · six more · reading the log · the RTX 3070 · a week-long run

BUILT ON

Running Persistent AI Worlds, second edition

SOURCES

7 numbered references

PREPARED FOR

Nerds489

About this paper

The research paper *Running Persistent AI Worlds* (second edition, 10 October 2026) ran two scenarios on a laptop: two learning agents in a forest that never ends (Scenario A), and an evolving population sealed in a box with a fixed amount of energy (Scenario B). This paper joins them into one world, adds a river, and sets it up to run for days on the Ryzen 9 5950X desktop.

- **A new simulation, tested.** `river_world.py` puts Scenario A's learning agents inside Scenario B's sealed, evolving population, on a cylinder of land split by a river that carries nutrients and can be crossed only at three fords. Energy is conserved and checked every tick; a stopped run resumes exactly where it left off (Appendix B).
- **A setup for sixteen cores.** One process per core, one seed per process, a launcher, a summariser and a map tool (Sections 3 to 5).
- **A first experiment, measured.** Does it matter whether a child inherits what its parent learned? Four seeds of each, 300 simulated years, run on the laptop (Section 6).
- **What to do next.** Six experiments, one change each, and how to read the log (Sections 7 and 8); the RTX 3070's place in all this (Section 9); and a checklist for a week-long run (Section 10).
- **The six experiments, run.** All six from Section 7, sixteen seeds each, 2,000 simulated years and 10,000 for the four still moving, on the desktop. Predictions written first and scored afterwards (Section 11).

Measured numbers in Sections 1 to 10 come from the laptop the paper was built on, and the desktop figures there are labelled as estimates. Section 11 was run on the desktop itself and replaces those estimates with measurements.

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30,000

energy units in the world, never created or destroyed, checked 400 times a simulated year

3 fords

the only way across the river; the world wraps north to south and is walled east to west

16 seeds

at once on the 5950X: one process per physical core, about 80 MB each

0.18 vs 0.05

learning rate after 300 years with learning inherited, and without: evolution turns learning down when it is not passed on (Section 6). By year 2,000 and sixteen seeds the gap is 0.06 against 0.03, and by year 10,000 0.039 against 0.033: the gap was a matter of speed, not destination (Section 11)

70 → 21

energy needed to reproduce, at the start and after 300 years: evolution chose many small, hungry agents (Section 6)

identical

a stopped and resumed run matches an uninterrupted one, byte for byte (Appendix B)

1 The desktop

This paper is written for one machine: the Ryzen desktop running Linux Mint 22.3. Its parts decide how the work is split. The simulation itself is NumPy on the CPU, one process per seed, so the sixteen cores are the engine. The graphics card is for the optional GPU stages in Section 9, and the memory and disks are simply never the constraint.

Table 1. The desktop and what each part does in this setup.

PART	SPECIFICATION	ROLE HERE
CPU	AMD Ryzen 9 5950X, 16 cores / 32 threads, boost to 4.9 GHz	Sixteen seeds at once, one per physical core
Memory	64 GB DDR4-3200 (2 × 32 GB, dual rank)	About 80 MB per seed: sixteen seeds use about 1.3 GB
GPU	NVIDIA GeForce RTX 3070 LHR, 8 GB	Optional: JAX environments and a local language model (Section 9)
Motherboard	Gigabyte B550M DS3H AC	
Operating system	Linux Mint 22.3 (Ubuntu 24.04 base), kernel 6.17	Python 3.12 from the distribution

Three consequences follow.

1. **Seeds, not threads.** One river world is a single-threaded Python loop over NumPy arrays. It cannot use sixteen cores for one world, but sixteen independent worlds use them perfectly. That suits the science too: a result that holds in one seed is an anecdote; one that holds in twelve of sixteen is a finding.
2. **Physical cores, not threads.** The 5950X shows 32 logical CPUs, but its second thread per core adds little to this kind of arithmetic. The launcher therefore runs `nproc / 2` = 16 processes, and pins NumPy to one thread each so they do not compete.
3. **Heat, not memory, is the limit for week-long runs.** Sixteen busy cores hold the 5950X at its all-core power limit for days. Watch `sensors` for the first hour (Section 5.3); the chip's limit is 90 °C.

> **Note:** Every timing in this paper was measured on the laptop the paper was built on (Intel Core i7-8665U, 4 cores, 16 GB). Figures for the desktop are estimates until you run the smoke test in Section 4, which takes a minute and replaces them with your own numbers.

2 River world: Scenarios A and B in one world

The research paper ran its two scenarios separately. Scenario A put two learning agents in a forest that never ends; Scenario B sealed an evolving population in a box with a fixed amount of energy. River world joins them: every agent learns during its life, as in A, inside a population that evolves under a fixed energy budget, as in B. A river runs down the middle.

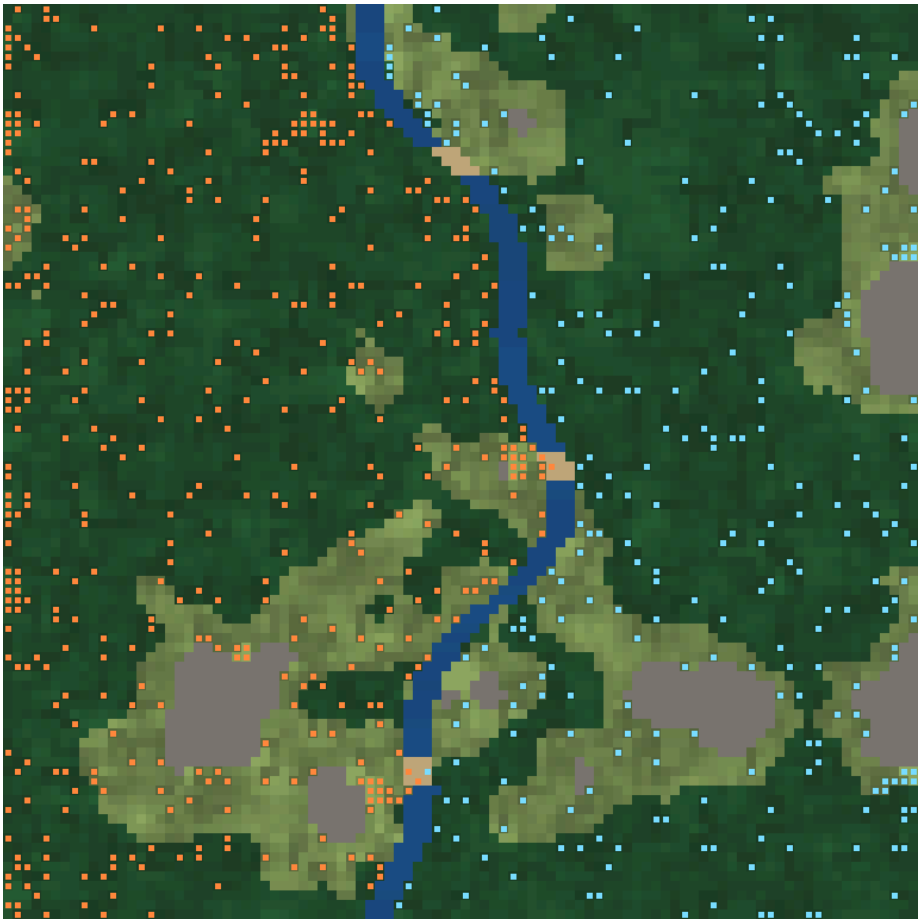


Figure 1. River world after 300 simulated years (seed 1, learning inherited). The river runs north to south and wraps round, so it has no source and no mouth; tan bars are the three fords. Agents are coloured by bank: west orange, east blue. Paler land holds more nutrients; the river darkens where it carries more silt.

2.1 What comes from Scenario A

- **Terrain** from the same smoothed-noise generator: forest, clearings (about 22% of the land) and rock (5%), which nothing can enter.
- **Time.** A day is 20 ticks, the last 40% of it night; a season is 100 ticks; a year is 400. Moving at night costs a quarter more.
- **A 2-bit signal channel.** Every action pairs a move (stay, north, east, south or west) with a signal (silent, 1, 2 or 3), 20 actions in all. An agent hears the loudest signal in the eight cells around it, one tick later. Speaking costs 0.05 energy.
- **Learning in a lifetime.** Each agent is a tabular Q-learner over 240 states: the direction of the richest neighbouring cell (5), its own energy band (3), day or night (2), the signal it hears (4) and whether it is within four cells of the water (2). The reward is the change in its own energy. Ties between untried actions are broken at random.

2.2 What comes from Scenario B

- **A sealed energy budget.** The world holds 30,000 energy units by default, and none is ever created or destroyed. The total is asserted after every tick.
- **Bodies and death.** Resting costs 0.5 a tick and moving 0.9; an agent eats up to 4 a tick from its cell and holds at most 150. It dies of starvation or of old age, and its body becomes detritus where it

falls.

- **Heritable genomes.** Five genes: lifespan, the energy needed to reproduce, the mutation scale, the learning rate and the exploration rate. A parent above its threshold splits its energy with a child whose genes are mutated. Evolution can therefore tune *how* agents learn, which is the Baldwin effect that Hinton and Nowlan demonstrated in 1987 [1].
- **What the child learns from the parent** is a switch, `--inherit`: `copy` hands over the parent's Q-table (learning passes down, Lamarckian in effect), `half` hands it over at half strength, and `none` starts every child blank, so only the genes for learning are inherited.

2.3 What the river adds

- **Two banks.** The world wraps north to south but has walls east and west, so it is a cylinder and the river cuts it in two. The channel is three cells wide and meanders up to about 11 cells either side of the centre line. Agents can cross only at three fords, at rows 16, 48 and 80.
- **A moving store of nutrients.** Each tick the river erodes 0.2% of the nutrients on its banks, carries a fifth of each stretch's load one row downstream, and lays 1% back down as silt on the banks. Because the river is a closed loop, everything it carries comes back.
- **Seasons and events.** Silt deposition rises two and a half times in spring. At the start of each season the world may roll an event: a spring flood (one year in four: six times the silt, three times the erosion, for 40 ticks), a summer drought (about one in seven: the river slows to a quarter and drops less silt), or a fire (a 12% chance at the start of each summer and autumn: a circle of radius 6 burns, nutrients become detritus, and agents caught in it lose half their energy). `--no-events` switches them off.

2.4 The energy ledger

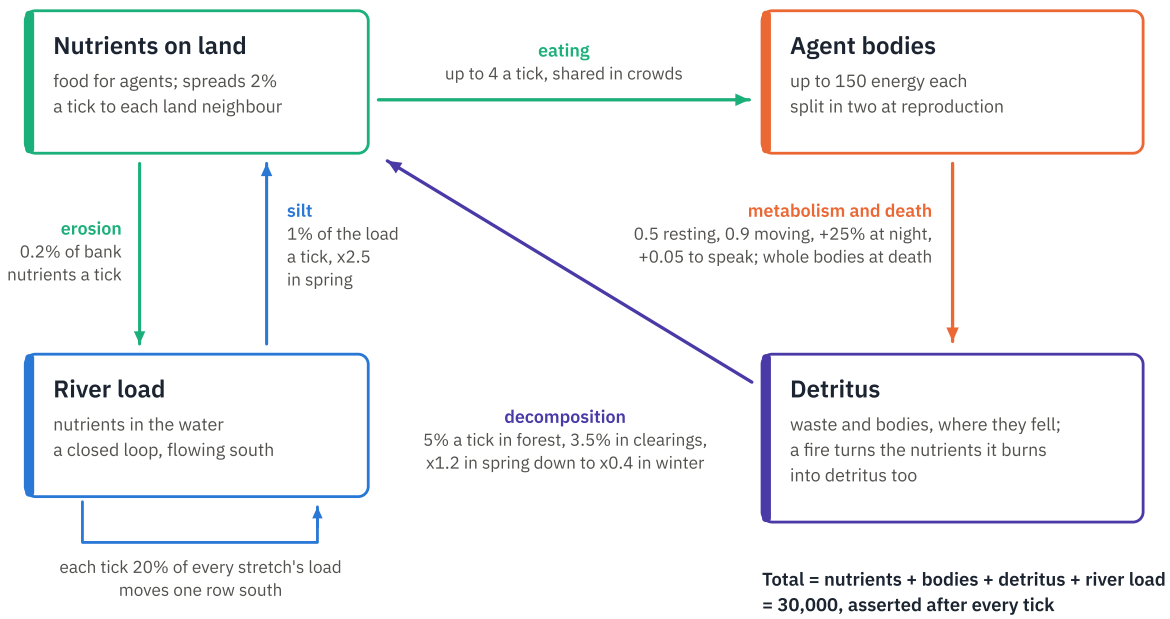


Figure 2. Where the energy goes. Four pools hold all 30,000 units; arrows are the only ways energy moves between them. Events change the rates on the arrows, never the total.

Decomposition turns detritus back into nutrients in place, at 5% a tick in forest, 3.5% in clearings and 2% on fords, and seasonally: a fifth faster in spring, under half as fast in winter. Nutrients also spread 2% a tick to each neighbouring land cell, never into rock or deep water. The full parameter list is in Appendix A.

> **Key insight:** Because nothing is created, every pattern in the population has to be paid for out of the same 30,000 units. Rich banks mean poorer land somewhere else; a boom on one side of the river is energy that is not available on the other.

3 Install

Everything in this section runs in a terminal on the desktop. The river world needs only Python 3.10 or newer and NumPy; the GPU stages in Section 9 have their own environments so that nothing here can break them.

3.1 The simulation environment

```
sudo apt update && sudo apt install -y python3.12-venv lm-sensors tmux
python3 -m venv ~/venvs/river && source ~/venvs/river/bin/activate
pip install -U pip numpy
python3 -c "import numpy, sys; print(sys.version.split()[0], numpy.__version__)"
```

Use the virtual environment rather than the system Python. On the laptop this paper was built on, a NumPy 2 in the user site directory shadowed the distribution's matplotlib, which was built for NumPy 1, and broke it; a venv keeps the two apart. Activate it (`source ~/venvs/river/bin/activate`) in every new terminal before running anything below.

3.2 The files

Copy the `river-world/` folder that comes with this paper to the desktop, for example to `~/river-world`. It holds five files and nothing else:

Table 2. The river world kit.

FILE	WHAT IT DOES
`river_world.py`	The world. One process runs one seed and writes `log.csv` and `checkpoint.pkl` in its run folder
`run_seeds.sh`	Runs many seeds at once, one per physical core, then summarises them
`summarise.py`	One line per seed, the mean and spread across seeds, and a comparison between conditions
`snapshot.py`	Draws a checkpoint as a PNG map (Figure 1 was made with it)
`README.md`	The commands from this paper, in short

```
cd ~/river-world && chmod +x *.py run_seeds.sh
python3 river_world.py --help
```

3.3 Check the graphics driver (only for Section 9)

```
nvidia-smi --query-gpu=name,driver_version,memory.total --format=csv
```

This should name the RTX 3070 with 8,192 MiB. Note the driver version: Section 9 installs the JAX build that matches it.

4 First run: four checks

Before committing sixteen cores to a week, spend ten minutes proving that the world works on this machine. Each check has a pass condition you can see.

4.1 Smoke test and speed

```
cd ~/river-world && source ~/venvs/river/bin/activate
time python3 river_world.py --years 20 --fresh --dir runs/smoke
```

It prints one line, for example:

```
runs/smoke: year 20  population 800 (west 393, east 407)  max generation 21  (0.99 years/s)  energy
conserved to 9.1e-10
```

Pass: the run finishes, both banks are populated, and `energy conserved to` is below 10^{-6} of the 30,000 total. On the laptop this paper was built on, one process ran 0.99 simulated years a second. Write down your own figure; Section 5 uses it.

4.2 Energy is conserved, every tick

There is nothing to do: the check is built in. After every tick the world adds up the nutrients, the detritus, the river's load and every body, and stops with `energy not conserved` if the total has drifted by more than one part in a million. A run that finishes has passed 400 checks a simulated year. The column `energy_drift` in `log.csv` records the drift at each year's end.

4.3 Stop, resume and get the same world

```
python3 river_world.py --years 6 --fresh --seed 5 --dir runs/straight
python3 river_world.py --years 20 --fresh --seed 5 --dir runs/stopped &
sleep 4; kill -INT %1; wait          # the same as pressing Ctrl-C
python3 river_world.py --years 6 --dir runs/stopped      # carries on from where it stopped
head -n 7 runs/straight/log.csv | cmp - <(head -n 7 runs/stopped/log.csv) && echo identical
```

Ctrl-C (or `kill`, or a shutdown) makes the world finish its current year, save, and exit. A checkpoint holds the whole world, including the random number generator, so a resumed run follows exactly the path an uninterrupted one would have. **Pass:** `identical`. If the machine loses power instead, the next start trims any log rows written after the last checkpoint (every 10 years by default; `--save-every 1` for every year) and carries on from there.

4.4 Look at it

```
python3 snapshot.py runs/smoke                    # agents coloured by bank
python3 snapshot.py runs/smoke --colour signal    # by each agent's favourite signal
xdg-open runs/smoke/map-year-20-side.png
```

Pass: a map like Figure 1: green land, a blue river down the middle, three tan fords, and agents on both banks.

5 Running sixteen seeds

5.1 The launcher

```
./run_seeds.sh 16 1000 main                          # 16 seeds x 1,000 years into runs/main/
./run_seeds.sh 16 1000 darwin --inherit none         # a second condition, its own folder
./run_seeds.sh 16 1000 main                          # again: every seed carries on another 1,000
years
```

The arguments are the number of seeds, the years to run, a name for the condition, and any `river_world.py` options. The launcher runs one process per physical core (16 on the 5950X), keeps NumPy single-threaded inside each, lowers their priority with `nice` so the desktop stays usable, and appends each seed's output to `runs/<name>/seed-N.out`. When every seed has finished it prints the summary from Section 6. Running it again resumes every seed; options such as `--inherit` only take effect when a seed's world is first created, and the world says so if you pass a different one later.

5.2 How long it takes

Table 3. Throughput, measured on the laptop and estimated for the desktop.

	LAPTOP (I7-8665U, MEASURED)	DESKTOP (5950X, ESTIMATE)
One process, about 800 agents	0.99 years/s	about 1.4 years/s
All cores busy	0.32 years/s per process, 8 processes on 4 cores	about 1.1 years/s per process, 16 processes
16 seeds × 1,000 years	about 1.7 hours (8 at a time)	about 15 minutes (all 16 at once)
16 seeds for a week		about 700,000 simulated years each

The desktop column assumes each 5950X core is about 1.4 times as fast as one of the laptop's on this single-threaded NumPy work, and that sixteen processes on sixteen cores lose about a fifth of that to shared memory bandwidth and lower all-core clocks. Treat it as a guess until Section 4.1 has given you the real one-process figure; the all-cores figure is the one to measure next, by running `./run_seeds.sh 16 20 speedtest` and reading the `years/s` on each line.

Population sets the pace: the work per tick grows with the number of agents. Doubling `--energy` roughly doubles the population and halves the speed.

5.3 Heat and power

A week of sixteen busy cores is a sustained load the 5950X will hold at its power limit. In the first hour:

```
watch -n 5 "sensors | grep -E 'Tctl|Tccd'"
```

Under 85 °C is comfortable; the chip throttles itself at 90 °C, which slows the runs but harms nothing. If it runs hot, use `JOBS=12 ./run_seeds.sh ...` to leave four cores idle. The runs do not care about the GPU, so it can be used for Section 9 at the same time.

5.4 Surviving the terminal

Run the launcher inside `tmux` so that closing the terminal or logging out does not stop it:

```
tmux new -s river           # start a session, then run ./run_seeds.sh inside it
# detach with Ctrl-b then d; reattach later with: tmux attach -t river
```

A reboot stops the runs cleanly (each one saves before it exits). Afterwards, run the same `./run_seeds.sh` line again and every seed resumes. Keep the kit and its `runs/` folder in your home directory, never under `/tmp`, which is emptied at every boot.

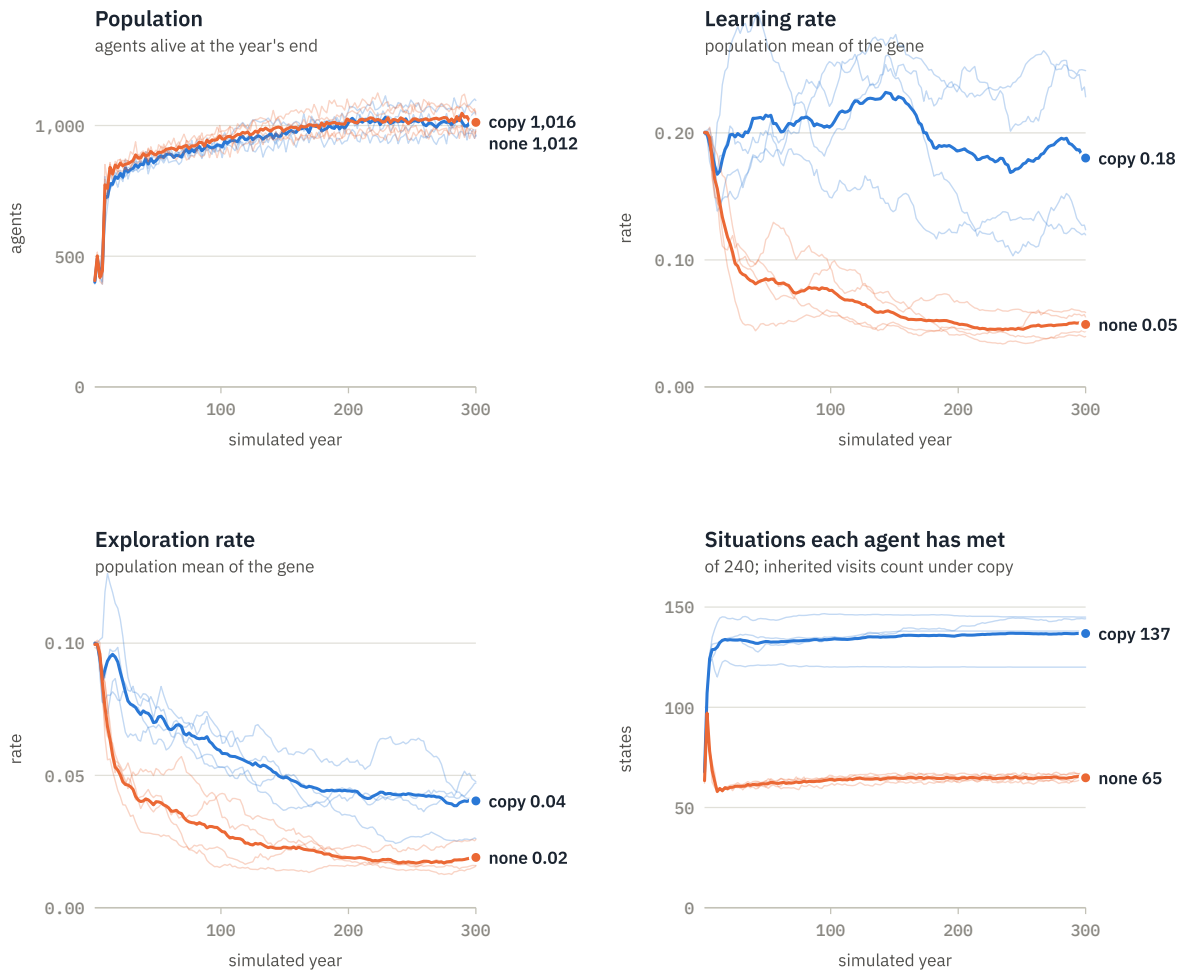
5.5 Disk

Each seed keeps one checkpoint of about 39 MB, overwritten in place (almost all of it is the 2,000 agent slots' Q-tables), and a log that grows by about 140 bytes a simulated year. Sixteen seeds take about 650 MB whatever the run length, plus their logs. To keep a snapshot of a world at some moment, copy its `checkpoint.pkl` somewhere else; it compresses to about a tenth with `gzip`. Checkpoints are Python pickles, and opening a pickle can run code, so only load checkpoints you made yourself.

6 The first experiment: should learning be inherited?

The one design choice river world forces on you is what a child gets from its parent's learning. The first experiment compares the two ends: `copy`, where a child starts with its parent's Q-table, and `none`, where it starts blank and inherits only the genes that set how it learns. Everything else was left at the defaults. Each condition ran four seeds for 300 simulated years on the laptop, eight processes at once, about 16 minutes in all.

```
./run_seeds.sh 4 300 inherit-copy
./run_seeds.sh 4 300 inherit-none --inherit none
python3 summarise.py runs/inherit-copy runs/inherit-none
```



Bold lines: the mean of four seeds; faint lines: each seed. copy = a child inherits its parent's Q-table; none = it starts blank.

Figure 3. The two conditions over 300 simulated years. Population ends the same, but the genes for learning go different ways: when learning is not inherited, evolution turns the learning rate and exploration down. Bold lines are the mean of four seeds; faint lines are the individual seeds.

Table 4. The last 50 years of each run, mean of four seeds (± standard error), with Welch's *t*.

MEASURE	COPY	NONE	*T*
Population	1,012 ± 24	1,024 ± 27	−0.3
Learning rate (gene, started at 0.2)	0.186 ± 0.033	0.049 ± 0.005	4.1
Exploration rate (gene, started at 0.1)	0.041 ± 0.005	0.018 ± 0.002	4.0
Situations met (of 240)	137 ± 6	65 ± 1	12.3
Deepest lineage (generations)	116 ± 7	77 ± 6	4.3
Signal entropy (bits; 2.0 = random)	1.961 ± 0.006	1.998 ± 0.000	−5.9
Behaviour diversity (`ecology_entropy`, bits)	6.07 ± 0.17	4.30 ± 0.17	7.4
Difference between the banks' signals (`dialect_jsd`)	0.0006	0.0001	5.6
Lifespan (gene, started at 1,200 ticks)	3,178 ± 257	3,213 ± 313	−0.1

6.1 What happened

Both worlds filled up the same way. Population rose from 300 to about 700 within ten years and settled near 1,000 in both conditions; the energy budget, not the inheritance rule, decides how many agents the river world feeds. Both populations also evolved the same strategy: the energy needed to reproduce fell from 70 to about 21, close to its floor of 20, and lifespans stretched from 1,200 to about 3,200 ticks. Evolution chose many small agents that split as soon as they can and then live long, hungry lives: the mean agent holds about half a unit of energy. Neither of those results was designed in.

The genes for learning went different ways. When children inherit their parents' Q-tables, the learning rate stays near where it started (0.19) and exploration falls slowly, from 0.10 to 0.04. When every child starts blank, evolution turns learning down: the learning rate falls steadily to 0.05 and exploration to 0.02. All four seeds agree. One reading is that a blank Q-table learned over a short, hungry life is mostly noise, and that noise costs energy, so selection quietens it; whereas an inherited table is worth updating carefully. That is the Baldwin effect's question, how learning and evolution shape each other [1], answered here in one specific way for one specific world. It is a result to test on sixteen seeds and longer runs (Section 7, E2), not a law.

Inherited learning carries more of the world. Agents under `copy` had met 137 of the 240 situations, twice as many as under `none`, because each table carries its ancestors' experience. They also kept a more varied population (6.1 bits of behaviour diversity against 4.3) and their lineages ran deeper.

No dialects yet. Signals stayed close to random in both conditions (entropy 1.96 and 2.00 bits out of 2), and the two banks used them almost identically: under `copy`, `dialect_jsd` peaked at 0.005 around year 25 and ended below 0.001. With three fords and about 300 crossings a year, 300 years is too short and too well connected for the banks to drift apart. That is exactly what Section 7's experiment E3 is for, on the desktop's sixteen cores and over thousands of years.

> **Note:** Four seeds a side is enough to see the large effects here (|t| above 4), but not to trust the small ones. The desktop runs sixteen seeds in the time the laptop ran four; run E1 and E2 there before building on any of this.

7 Experiments to run next

Each experiment below is one change against the defaults, run on 16 seeds. Before you start one, write your prediction in a file next to the runs; the paper's first habit applies here more than anywhere: emergence only means something against a prediction.

Table 5. Six experiments for the desktop, each a single change.

#	QUESTION	COMMAND (16 SEEDS EACH)	READ THESE COLUMNS
E1	Does handing learning down help?	`./run_seeds.sh 16 2000 copy` and `./run_seeds.sh 16 2000 none --inherit none` (add `half` as a third)	`population`, `q_states`, `policy_complexity`
E2	Does evolution tune how agents learn (the Baldwin effect [1])?	The same two conditions as E1	`gene_alpha`, `gene_eps`: do they drift apart between conditions?
E3	Is the river a barrier to culture?	Defaults, 5,000 years	`dialect_jsd`, `crossings`, `pop_west`, `pop_east`
E4	Does disturbance keep the world interesting?	`./run_seeds.sh 16 2000 calm --no-events` against the defaults	`novelty`, `ecology_entropy`, `change`
E5	Starvation or crowding?	`--energy 20000`, the default 30,000, and `--energy 45000`	`population`, `gene_repro`, `gene_lifespan`, `mean_energy`
E6	Does a bigger world sustain more variety?	`--size 128 --energy 53000` (the same energy per cell)	`archive_size`, `ecology_entropy`, `dialect_jsd`

Two experiments need a one-line edit rather than an option, and the code is short enough to make it safely. To change how many fords there are, edit the `self.fords = [...]` line in `World.__init__`; to make the river a wall, give it no fords at all. Make the edit in a copy of the folder, so that the original runs keep resuming with the code they started with.

> **Caution:** A change to the code applies to every world that resumes under it. Never edit `river_world.py` under a running experiment: copy the folder, change the copy, start fresh worlds.

7.1 The three habits, applied

1. **Predict first.** For E3, for example: "with three fords, `dialect_jsd` stays under 0.05; with one ford it climbs above 0.2 within 2,000 years." A number you can be wrong about.
2. **Change one thing, run sixteen seeds.** `summarise.py runs/A runs/B` prints a Welch *t* for every column. With 16 seeds a side, |t| above about 2 is worth a closer look; below that, the difference is within what the seeds do on their own.
3. **Keep everything.** Copy a seed's `checkpoint.pkl` when something odd happens, and take a `snapshot.py` map. The interesting moment in a long run is usually found afterwards, in the log.

7.2 Measuring open-endedness, honestly

The log carries simple versions of the four MODES measures (change, novelty, complexity and ecology) [2]. They are deliberately crude: each agent is summarised by a six-part behaviour descriptor (learning-rate band, exploration band, reproduction band, favourite signal, mobility band and bank), and `novelty` counts descriptors never seen before in that run. The descriptor space has 2,048 cells, so novelty is bound to fall to zero as a run fills it; that is a property of the measure, not proof that the world has stopped producing anything. Whether any of this counts as open-ended depends on the observer [3], and here the observer is these six numbers. Richer descriptors are the obvious first improvement.

8 Reading the log

Each run writes one row a simulated year to `log.csv`. Open it in any spreadsheet, or read it with `summarise.py`.

Table 6. The columns of `log.csv`.

COLUMN	MEANING
`year`	Simulated years since the world was created (400 ticks each)
`population`, `pop_west`, `pop_east`	Agents alive at the year's end, in total and on each bank
`crossings`	Times agents changed bank during the year (only possible at fords)
`mean_energy`	Mean energy of a living agent (out of 150)
`nutrients`, `detritus`, `river_load`	Energy in each pool; with the agents' bodies they always sum to the total
`max_generation`	The deepest lineage alive: generations since the first 300 agents
`gene_lifespan`, `gene_repro`, `gene_mutation`, `gene_alpha`, `gene_eps`	Population means of the five genes: lifespan in ticks, energy to reproduce, mutation scale, learning rate, exploration rate
`q_states`	Mean number of the 240 situations an agent's Q-table has visited (inherited visits count under `--inherit copy`)
`policy_complexity`	Mean number of different actions an agent would choose across the situations it has visited (1 to 20)
`signal_entropy`	Entropy of all signals used in the year, in bits (2.0 means all four equally often)
`dialect_jsd`	Jensen-Shannon divergence between the two banks' signal use, in bits (0 means identical, 1 means disjoint)
`novelty`	Behaviour descriptors seen for the first time this year
`ecology_entropy`	Entropy of the population across behaviour descriptors, in bits
`change`	How far the mean genome has moved in ten years, in units of each gene's mutation scale
`archive_size`	Distinct behaviour descriptors seen since the world began (at most 2,048)
`energy_drift`	Total energy minus 30,000 at the year's end: rounding error only
`event`	Floods, droughts and fires that started during the year

9 Using the RTX 3070

River world never touches the GPU, so the card is free for the two GPU steps of the research paper's eight-week path: fast JAX environments, and a local language model. Give each its own virtual environment.

9.1 JAX on the GPU

The JAX build has to match the driver that Section 3.3 reported. `jax[cuda13]` needs driver 580 or newer, and `jax[cuda12]` needs 525 or newer; the RTX 3070 (Ampere) is supported by both.

```
python3 -m venv ~/venvs/jax && source ~/venvs/jax/bin/activate && pip install -U pip
pip install -U "jax[cuda13]"        # driver 580 or newer; otherwise: pip install -U "jax[cuda12]"
python3 -c "import jax; print(jax.devices())"          # should list a CUDA device, not only the
CPU
pip install jaxmarl craftax
```

JaxMARL [4] and Craftax [5] run whole populations of environments on the GPU at once; follow each project's README for a first training run. With 8 GB of memory, start with their default batch sizes and halve them if JAX reports that it is out of memory.

9.2 A local language model

```
curl -fsSL https://ollama.com/install.sh | sh
ollama pull qwen3.5:4b                 # 3.3 GB; qwen3.5:2b is 2.7 GB
ollama run qwen3.5:4b "In one sentence, what is a ford?"
```

Qwen3.5's 4B model fits in the 3070's 8 GB with room for a working context; the 9B model (6.6 GB) fits too, but leaves little room [6, 7]. Watch `nvidia-smi` while it answers.

9.3 Why language models cannot run river world's agents

A river world of 800 agents takes 800 decisions a tick and 320,000 a simulated year. Even at ten model calls a second, one simulated year would take about nine hours, and a 1,000-year run about a year. The Q-learners decide in microseconds. Language-model agents belong in a different design: a handful of them, observing the river world or advising one bank, while the population underneath stays cheap. That is step 7 of the research paper's eight-week path, and its Section 11 checklist applies the moment a language model is in the loop: log every prompt, and keep the model's access to the world read-only until you know what it does.

10 Before a week-long run

1. **Prediction written down** in `runs/<name>/PREDICTION.md`, with numbers.
2. **Smoke test passed** on this machine today (Section 4), including the resume check.
3. **One change per condition**, each condition in its own folder, 16 seeds each.
4. **Launcher inside** `tmux` (Section 5.4), so a closed terminal does not stop it.
5. **Temperatures watched for the first hour** (Section 5.3); `JOBS=12` if the CPU runs hot.
6. **Disk checked:** `df -h ~` shows at least 2 GB free for 16 seeds.
7. **The code frozen:** no edits to `river_world.py` while its worlds are running (Section 7).
8. **Summarise daily:** `python3 summarise.py runs/A runs/B` takes a second and changes nothing.
9. **Snapshot weekly,** or whenever a column does something odd: `python3 snapshot.py runs/A/seed-3`.

10. **Copy any checkpoint worth keeping** before the run overwrites it.
11. **After a reboot,** run the same `./run_seeds.sh` line again: every seed resumes.
12. **At the end,** keep the logs, the summaries and the prediction together, and write one paragraph on what was right and what was wrong.

11 The six experiments, run

Section 7 set out six experiments and asked for a prediction before each one. This section runs all six on the desktop and keeps that bargain. Twenty-seven numbered predictions with explicit falsifiers were written into `_build/experiment-s7/predictions.md` and timestamped before the first world existed, and two more blocks were added before later runs began; Section 11.8 scores all of them without editing them, and two were dead before the first condition finished.

Nine conditions, sixteen seeds each, one change against the defaults. Section 6 ran four seeds for 300 years on the laptop; this is sixteen seeds for 2,000 years, with the four conditions that were still moving or still unanswered carried on to 10,000 (Section 11.7), on the machine the paper was written for. In all, 800,000 seed-years.

Table 7. The nine conditions. Anything not named is at the stock default: 96 × 96 cells, 30,000 energy units, a 2,000-agent cap, learning inherited in full, events on.

CONDITION	CHANGE	EXPERIMENTS IT SERVES	YEARS
`base`	none	E1, E2, E4, E5 baseline; E3's three-ford arm	10,000
`none`	`--inherit none`	E1, E2	10,000
`half`	`--inherit half`	E1, E2	2,000
`calm`	`--no-events`	E4	2,000
`e20k`	`--energy 20000`	E5	2,000
`e45k`	`--energy 45000 --capacity 3000`	E5	2,000
`big`	`--size 128 --energy 53000 --capacity 3500`	E6	2,000
`ford1`	one ford instead of three	E3	10,000
`wall`	no fords at all	E3	10,000

Two notes on that table. The agent cap is a guard rail rather than a treatment: it sits at about twice the expected population in `base`, and it is scaled with the energy in `e45k` and `big` so that it stays equally far from binding instead of quietly capping one arm and not the others. And the two river arms run from copied folders, `_build/experiment-s7-ford1` and `-wall`, each differing from the kit by exactly one line, as Section 7 instructs; their terrain was checked before they ran, giving 27 ford cells with three fords, 9 with one and 0 with none.

11.1 The desktop, measured

Section 5.2 estimated the desktop's throughput from the laptop's and said to treat it as a guess. It was a good guess.

Table 8. Throughput: the Section 5.2 estimate against sixteen seeds actually run.

	LAPTOP, MEASURED	DESKTOP, ESTIMATED IN SECTION 5.2	DESKTOP, MEASURED HERE
All cores busy, one condition	0.32 years/s per process, 8 processes	about 1.1 years/s per process, 16 processes	**1.095 years/s per process** (range 1.06 to 1.15), 17.5 years/s in total
Two conditions at once, 32 processes		not estimated	0.630 years/s per process, **20.15 years/s in total**
16 seeds × 1,000 years	about 1.7 hours	about 15 minutes	**15.2 minutes**
16 seeds × 2,000 years			31 minutes 23 seconds, wall clock
16 seeds for a week		about 700,000 simulated years each	about 662,000
Temperature, all cores busy		under 85 °C wanted	49 to 51 °C

The estimate was good to 0.5% on the figure that matters. The chip never came close to needing `JOBS=12`: sixteen busy cores held it at about 50 °C, 35 degrees under the comfort line in Section 5.3, and thirty-two held it at the same temperature. Memory was never a factor either.

The second thread is worth 14%, which is more than "little". Section 5.1 sets `JOBS` to `nproc / 2` on the grounds that "the second thread of each core adds little to NumPy work like this". Measured directly, with two conditions running and one of them frozen with `SIGSTOP` for the comparison window rather than killed, over matched 180-second windows at the same simulated years:

PROCESSES	AGGREGATE
16, one condition	17.688 years/s
32, two conditions	20.153 years/s
64, four conditions	15.56 years/s

A gain of 1.139. The direction of Section 5.1's claim is right, since doubling the processes buys 14% rather than anything like 100%, and any single condition still finishes fastest on sixteen. But the practical reading is different from the one the launcher implies. A condition has only sixteen seeds, so the choice is never "16 jobs or 32" for one condition; it is whether to run a second condition alongside it or queue it behind. **Two conditions run together finish 14% sooner than the same two run one after the other**, at the cost of each individual seed taking 43% longer. Over a week that is most of a day. The experiment programme in this section used it: `ford1` and `wall` were run side by side.

And past the thread count it goes backwards. Four conditions at once, 64 processes on 32 threads, ran at 15.56 years a second: slower than 32 processes by a quarter and slower even than sixteen. Each process carries about 77 MB of Q-tables, so 64 of them ask the caches and memory bus for five gigabytes of working set at once, and time-slicing makes it worse. The rule for this machine is one process per logical thread, never more: two conditions at a time, and a queue for the rest.

The world is reproducible across machines. Section 6's runs were made on the laptop, and their own logs say "4 at a time", which is `nproc / 2` on a 4-core i7-8665U. Running `base` here regenerates seeds 1 to 4 of that same condition from scratch on the 5950X, and for all 300 years the two machines agree on every one of the 24 columns of `log.csv`, byte for byte, with a single exception: `energy_drift`, the floating-point residue of the conservation check, which differs in its last digits and reaches the same 9.78×10^{-9} maximum. A seed is a world, on either machine. That is what makes the rest of this section legitimate rather than merely new: the 2,000-year arms here begin by reproducing the 300-year arms there, and the Section 6 figures can be read as the first 300 years of Figure 4.

One correction while the machine is in view: Table 1 gives the kernel as 6.17, and this desktop runs 7.0.0-38-generic. Mint 22.3 and Python 3.12 are right.

11.2 E1 and E2: should learning be inherited?

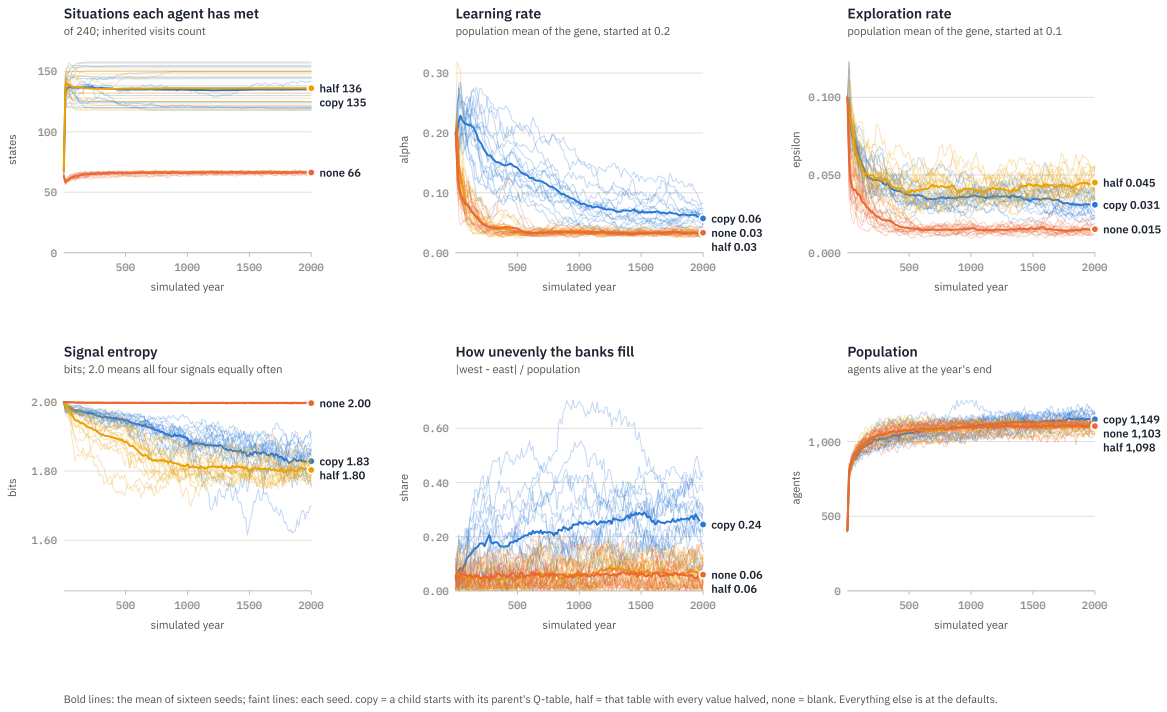


Figure 4. Three inheritance rules over 2,000 simulated years, sixteen seeds each. Bold lines are the mean of the sixteen; faint lines are the individual seeds. The learning-rate gap that Section 6 measured at year 300 is real, and it is narrowing. The middle rule is not a middle world.

Table 9. The last 100 years of each arm at year 2,000, mean of sixteen seeds (± standard error), with Welch's *t* for `copy` against `none`.

MEASURE	COPY	HALF	NONE	*T*
Population	1,151 ± 9	1,102 ± 6	1,105 ± 6	4.2
Learning rate (gene, started at 0.2)	0.061 ± 0.004	0.033 ± 0.001	0.033 ± 0.001	6.4
Exploration rate (gene, started at 0.1)	0.031 ± 0.002	0.045 ± 0.002	0.015 ± 0.001	9.2
Situations met (of 240)	135.1 ± 3.6	136.1 ± 3.5	66.4 ± 0.2	19.0
Policy complexity (of 20)	19.78 ± 0.04	18.96 ± 0.01	18.14 ± 0.01	41.0
Deepest lineage (generations)	540 ± 16	461 ± 9	382 ± 8	8.6

MEASURE	`COPY`	`HALF`	`NONE`	*T*
Signal entropy (bits; 2.0 = random)	1.828 ± 0.013	1.806 ± 0.006	1.9968 ± 0.0001	−13.3
Bank imbalance (|west − east| / population)	0.268 ± 0.027	0.065 ± 0.012	0.053 ± 0.012	7.3
Difference between the banks' signals (`dialect_jsd`)	0.0060 ± 0.0018	0.00046 ± 0.00007	0.00016 ± 0.00001	3.2
Behaviour diversity (`ecology_entropy`, bits)	4.81 ± 0.09	3.95 ± 0.04	4.12 ± 0.04	6.7
Crossings a year	232 ± 16	280 ± 9	319 ± 17	−3.7
Lifespan (gene, started at 1,200 ticks)	3,712 ± 48	3,891 ± 26	3,888 ± 41	−2.8
Energy to reproduce (gene, started at 70)	21.41 ± 0.02	21.22 ± 0.01	21.17 ± 0.01	13.5
Distinct behaviours ever seen (of 2,048)	751 ± 31	672 ± 34	671 ± 33	1.8

The Section 6 result survives, and shrinks. Learning is still turned down when it is not passed on, at sixteen seeds and 2,000 years, with $t = 6.4$. But the size of the effect is a property of when you look at it. Both arms fall; the inherited one just falls later:

SIMULATED YEAR	10	50	100	300	500	1,000	1,500	2,000
`copy`	0.195	0.222	0.214	0.157	0.143	0.084	0.068	0.057
`none`	0.174	0.094	0.080	0.042	0.041	0.034	0.032	0.033
ratio	1.1	2.4	2.7	3.7	3.5	2.5	2.1	1.7

The ratio peaks near year 300, which is exactly where Section 6 stopped, and decays to 1.7 by year 2,000. The front page of this paper quotes "0.18 vs 0.05 after 300 years"; that is true and it is the high-water mark. At 2,000 years the inheriting world's learning rate, 0.057, has fallen to about where the blank-slate world's was at year 300. Section 6 called the result "a result to test on sixteen seeds and longer runs, not a law", and this is what testing it did: the direction held, the magnitude did not. Carried on to year 10,000 (Section 11.7), the ratio keeps falling, to 1.20, and drops inside seed noise.

A halfway rule is not a halfway world. `half` was added as E1's third arm on the assumption that it would land between the other two. It does not land anywhere consistent. It matches `copy` on situations met ($t = -0.20$) and on signal entropy ($t = 1.6$); it matches `none` on the learning-rate gene ($t = 0.17$) and on bank imbalance ($t = 0.69$); and on exploration and behaviour diversity it is outside the range of both, with the highest exploration of the three arms and the lowest diversity. Only policy complexity and lineage depth order the three arms monotonically.

The reason is in one line of `river_world.py`:

```
elif self.inherit == "half":
    self.Q[ch], self.visited[ch] = 0.5 * self.Q[p], self.visited[p]
```

`half` does not give a child half of its parent's table. It gives the whole visited mask and every value halved, and the halving compounds down the generations. Reading all forty-eight year-2,000 checkpoints:

Table 10. What an agent actually carries at year 2,000, mean of sixteen seeds (± standard error).

ARM	VISITED STATES (OF 240)	MEAN |Q| OVER VISITED STATES	MEAN BEST Q	SHARE OF VISITED CELLS STILL EXACTLY 0
`copy`	135.1 ± 3.6	0.4476 ± 0.018	3.044 ± 0.21	0.360 ± 0.017
`half`	136.1 ± 3.5	0.00390 ± 0.00016	0.4469 ± 0.008	0.408 ± 0.015
`none`	66.35 ± 0.21	0.00617 ± 0.00016	0.3428 ± 0.0061	0.698 ± 0.001

A `half` agent claims to have met more situations than a `copy` agent and holds values 115 times weaker, weaker even than what a blank-slate agent learns for itself in one life. Halving preserves which action is best, so the parent's policy does transmit, which is why signalling still becomes structured; it does not preserve how strongly that action is preferred, and after hundreds of generations of halving the preference is worth less than a single reward. Evolution reads that correctly and tunes the learning rate as if nothing were inherited.

> **`q_states` is not a measure of inherited knowledge.** Table 6 notes that inherited visits count under `--inherit copy`. They count under `half` too, and there the flag outlives the knowledge entirely. Anyone using `q_states` to compare inheritance rules should read Table 10 first, or use `policy_complexity`, which does order the three arms.

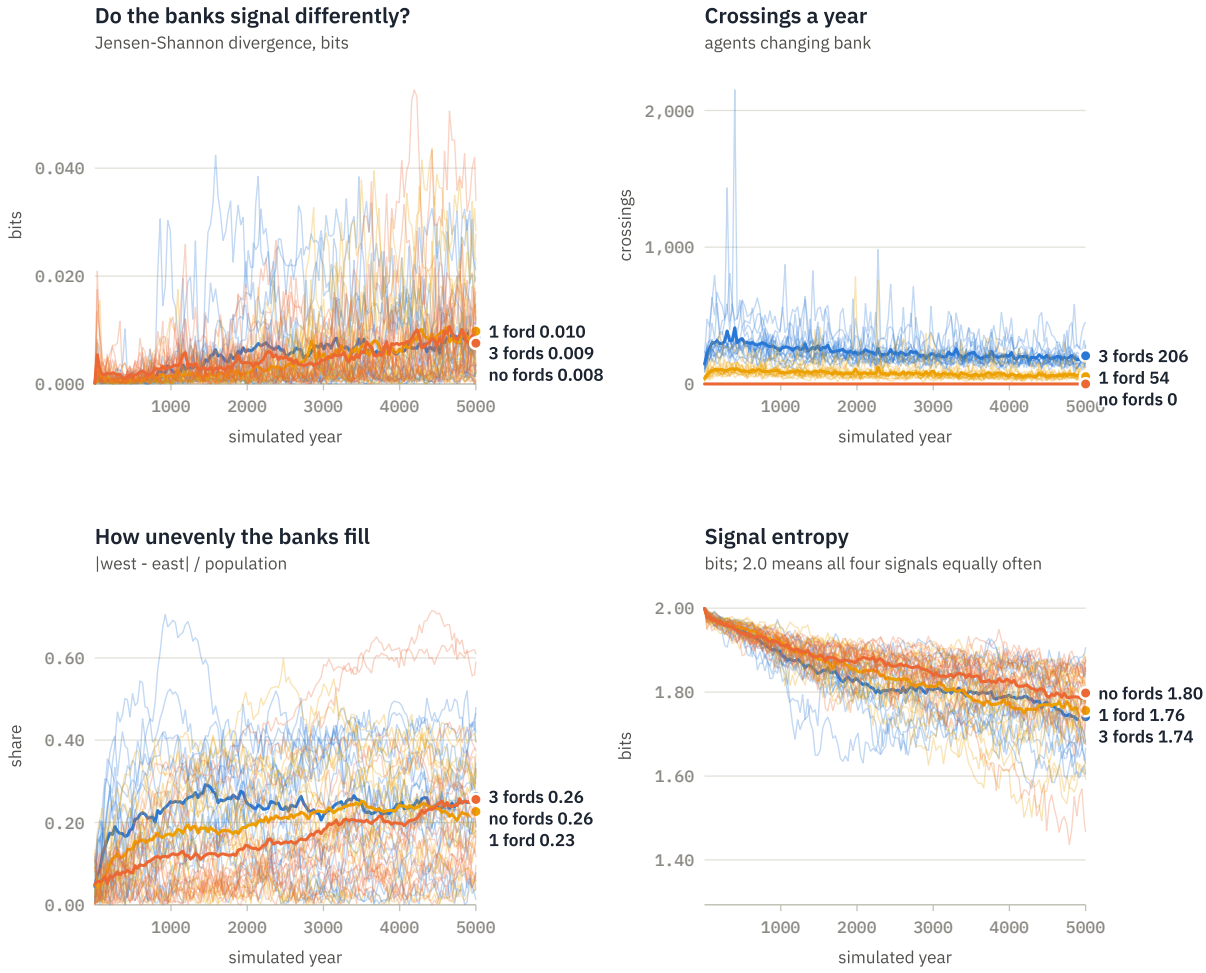
Section 6's table replicates at sixteen seeds. Read at year 300, the arms here give a learning rate of 0.162 ± 0.013 against 0.043 ± 0.002, where four seeds gave 0.186 ± 0.033 against 0.049 ± 0.005; and situations met of 136.2 against 65.3, where four seeds gave 137 against 65. The four-seed numbers were right; sixteen seeds mostly tightened them, and lifted t on the learning rate from 4.1 to 8.9.

Two things Section 6 did not report, and both grow with time. Neither is new at year 2,000, and that is the point: each was already measurable at year 300 and each is several times larger by the end.

- **Inherited learning makes signalling structured.** Signal entropy under `copy` falls from 1.9598 bits at year 300 to 1.828 at year 2,000, while `none` sits at 1.9976 and then 1.9968, four ten-thousandths off a perfectly random 2.0 and never moving. The gap widens from 0.038 bits to 0.169, with t going from −9.6 to −13.3. A blank-slate world never leaves random signalling, because nothing about how to signal survives its user.
- **Inherited learning makes the banks fill unevenly.** The imbalance between the banks under `copy` runs 0.187 of the population at year 300 and 0.268 at year 2,000, against 0.049 and 0.053 under `none`, t = 4.4 then 7.3. Which bank wins is seed-dependent (seed 7 ends 845 west to 327 east, seed 12 ends 418 to 773), and nothing in the code favours a bank. It is the compounding of one bank's inherited tables, and it is absent in both arms where inheritance is effectively switched off.

Finally, a result that only sixteen seeds could reach: population does separate. 1,151 against 1,105, t = 4.2. Four seeds a side could not see a 4% difference and Section 6 reported none; the energy budget still sets the order of magnitude, but inherited learning is worth about forty agents.

11.3 E3: is the river a barrier to culture?



Bold lines: the mean of sixteen seeds; faint lines: each seed. The arms differ by one line of code: how many fords cross the river.

Figure 5. The river experiment, sixteen seeds an arm over 5,000 simulated years. The three worlds differ by one line of code, the number of fords. Crossings fall from about 200 a year to about 60 to none, and the difference between the banks' signalling does not move with them.

No. Cutting the river off completely leaves the two banks signalling exactly as alike as they do with three fords open.

The three arms are the defaults with three fords (`base`), one ford (`ford1`) and none (`wall`), each a one-line edit of `self.fords` in a copied folder, sixteen seeds a side, scored at year 5,000 as pre-registered.

Table 11. E3 at year 5,000: the mean over years 4,901 to 5,000, sixteen seeds an arm, with Welch's *t* against `base`.

	THREE FORDS	ONE FORD	*T*	NO FORDS	*T*
Crossings a year	193	62	7.7	0	13.1
`dialect_jsd`, bits	0.0086	0.0090	−0.15	0.0083	0.12
Bank imbalance, |west − east| / population	0.253	0.216	0.85	0.255	−0.03
Seeds with imbalance above 0.25	8 of 16	7 of 16		6 of 16	
Signal entropy, bits	1.739	1.753	−0.52	1.790	−1.72
Behaviour diversity, bits	4.27	4.27	0.04	4.48	−1.09
Population	1,177	1,198	−1.46	1,180	−0.22
Smallest bank in any year, any seed	153	135		141	

The treatment worked as built: the walled world logged zero crossings in every seed and every one of its 80,000 seed-years, and one ford cut traffic to a third. Nothing downstream of the treatment followed it. The between-bank divergence is 0.0086, 0.0090 and 0.0083 bits, $t = 0.12$ for complete isolation against open fords. The arms never order the way contact would predict. Checked every 500 years from 2,000 on, the walled world is the highest of the three at one of the seven checkpoints (4,000) and the lowest at three (3,000, 3,500 and 5,000); the one-ford world is the lowest at 2,000, 2,500 and 4,000 and the highest at 4,500 and 5,000. Every arm drifts upward together, from about 0.001 at year 500 to about 0.009 at 5,000, which is a property of the world ageing, not of its geography.

The interim reading at year 2,000 had found the connected world's banks differing *more* than the isolated ones, and read it as one bank's inheriting lineage dominating and spilling across the fords. By 5,000 that difference has gone and so has the link: across all 48 seeds, a seed's bank imbalance and its divergence are uncorrelated ($r = -0.01$), and so are its divergence and the size of its smaller bank ($r = 0.07$). The interim gap was seed noise of the kind 16 seeds a side produces at $t = 2$.

Why there is nothing for isolation to separate. The answer is in what the agents say, and mostly it is nothing. Reading the final world state of every seed, the agent's favourite signal over its last year is silence for 79 to 80% of agents on both banks with three fords open, and 75 to 76% on both banks with none. The other three symbols split the rest almost evenly, 5 to 9% each, on both banks of both worlds. Speaking costs 0.05 energy a tick whichever symbol is used and silence costs nothing, so selection pushes every population towards silence identically, and among the three spoken symbols there is nothing to choose: they cost the same and the model gives no one of them a meaning the others lack. A dialect needs an arbitrary convention for two isolated populations to settle on differently. This channel has a cost gradient instead, and a cost gradient points the same way on both banks. Signal entropy falls steadily in every arm (from 2.0 at year 1 to about 1.75 at 5,000) as the channel goes quiet, and it falls slowest in the walled world, $t = -1.7$ by 5,000: if anything, contact speeds the drift into silence.

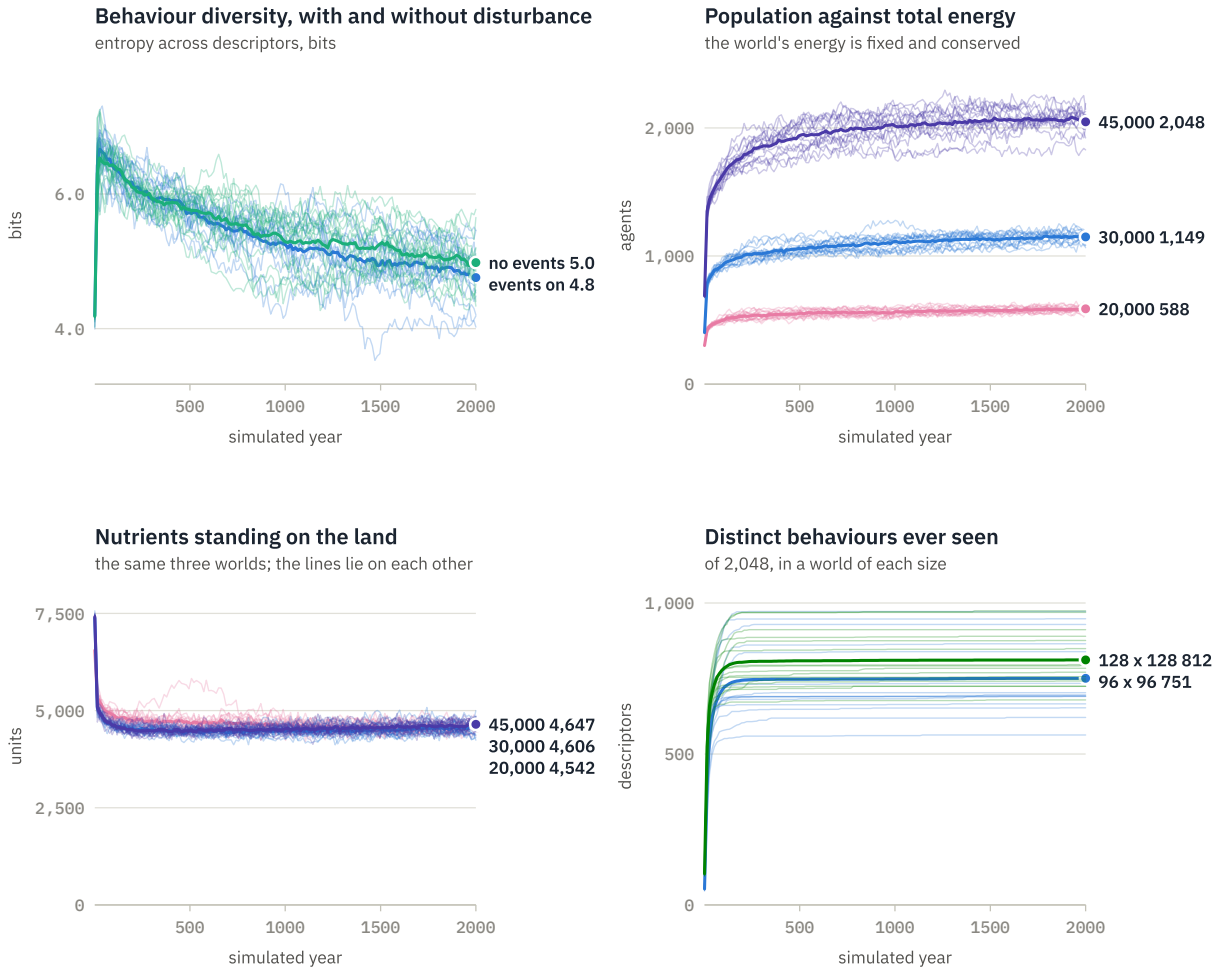
What `dialect_jsd` measures in this world, then, is the sampling wobble of two near-identical, mostly silent distributions. The largest value any seed reached in any single year of any arm was 0.074, and that in the walled world; the paper's threshold of 0.2 was never approached.

Against the paper's own worked example. Section 7 predicted "with three fords `dialect_jsd` stays under 0.05; with one ford it climbs above 0.2 within 2,000 years". The first half holds by a factor of six. The second is wrong by a factor of a hundred: one ford stood at 0.0020 at year 2,000, below the three-

ford world, and its single highest seed-year in 5,000 years was 0.052. The pre-registered falsifier for the river as a barrier, "`wall` staying below 0.01: signals are driven by terrain and season, not by contact", is triggered at 0.0083, and the reading it names is the right one, sharpened: signals here are driven by their cost.

The experiment worth running next is the one that gives the channel something to diverge on. Make the three spoken symbols carry different information (one heard as "food here", say, by tying it to a nutrient threshold in `observe`) or make a symbol's use pay only when a neighbour answers in kind, and then compare `wall` against `base`. Until the model has a convention to break, isolating its banks cannot break one. Whether ten thousand years of drift changes that verdict for the river arms is in Section 11.7.

11.4 E4: does disturbance keep the world interesting?



Bold lines: the mean of sixteen seeds; faint lines: each seed. Each panel changes one thing against the defaults: whether fires, floods and droughts happen; how much energy the world holds; how big it is.

Figure 6. Disturbance, energy and size, sixteen seeds an arm over 2,000 simulated years. Top left: switching fires, floods and droughts off changes nothing measurable (11.4). Top right and bottom left: population tracks total energy more than proportionally while the nutrients standing on the land do not move at all (11.5). Bottom right: a world nearly twice the size explores barely more behaviours (11.6).

No. Not as the three disturbances are currently sized.

`calm` is the defaults with `--no-events`. Against `base` at year 2,000, sixteen seeds a side, every measure in the log falls inside seed noise. The largest $|t|$ among the twenty-two columns is 1.9, and the two that come closest both point the wrong way for the hypothesis: the calm world has slightly *more* behaviour diversity (5.00 against 4.81 bits, $t = -1.6$) and slightly *more* structured signalling (1.859 against 1.828 bits, $t = -1.9$). Population, lineage depth, the five genes, the archive and `change` are all flat. The only column that separates is the treatment itself: 1,037 events in 2,000 years against 0.

That is a negative result, so it is worth showing that the instrument was not asleep. Two checks inside `base` itself:

- **Years with an event look like years without.** Over years 500 to 2,000 and sixteen seeds, the annual change in population averages +0.060% in the 12,485 year-slots carrying an event and −0.029% in the 11,515 that do not, with standard deviations of 1.540 and 1.536.
- **Events add no volatility.** The standard deviation of the annual change, `base` against `calm`: population 1.537 against 1.521, nutrients 2.287 against 2.252, detritus 0.417 against 0.406, mean energy 6.407 against 6.291. Ratios of 1.01 to 1.03. The two columns that move at all, `q_states` at 1.19 and signal entropy at 1.11, are the smallest numbers on the list in absolute terms.

The reason is arithmetic, and it is in the pools the events touch. A fire burns a disc of radius 6, so about 113 of the world's 8,494 land cells. It moves the nutrients under that disc into detritus, about 60 of the 4,544 units on land, which decay back at 5% a tick; and it takes half the body energy of the agents standing in it, about fifteen agents holding about 0.45 each, so roughly 3 units. Floods and droughts change erosion, flow and deposition for 40 and 100 ticks, and all three of those act on the river's load, which holds **24 units of the world's 30,000**. Every disturbance in the model is working on 0.1% of the world's energy, in a world where 25,000 of the 30,000 units sit in detritus and are rained back onto the land continuously.

So E4's question is not answered by this run; it is deferred until the disturbances are large enough to answer it. A rerun worth doing on the desktop: scale fire's radius with the world (radius 20 burns 12% of the land rather than 1.3%), make drought suppress decomposition rather than river flow, since decomposition is the pipe the whole world feeds through, and only then compare against `calm`. The prediction to write first is the one this run can already falsify: that a disturbance confined to 0.1% of the energy budget cannot move a population of 1,100.

11.5 E5: starvation or crowding?

Neither. The land is grazed to the same standing crop whatever the world contains, and everything else is stored as detritus.

Table 12. Three worlds differing only in how much energy they hold, at year 2,000, mean of sixteen seeds (± standard error), with Welch's *t* against `base`.

MEASURE	`E20K` (20,000)	*T*	`BASE` (30,000)	`E45K` (45,000)	*T*
Population	584 ± 5	54.5	1,151 ± 9	2,066 ± 22	−39.2
Nutrients on land	4,558 ± 18	−0.4	4,544 ± 33	4,610 ± 29	−1.5
Standing crop a land cell	0.5350		0.5334	0.5411	
Detritus	15,133 ± 18	224.5	24,913 ± 40	39,490 ± 33	−280.7

MEASURE	E20K (20,000)	*T*	BASE (30,000)	E45K (45,000)	*T*
Energy in bodies	286		519	872	
`mean_energy`	0.489 ± 0.004	−4.9	0.451 ± 0.007	0.422 ± 0.004	3.7
Energy to reproduce (gene)	21.28 ± 0.01	5.6	21.41 ± 0.02	21.48 ± 0.02	−2.8
Learning rate (gene)	0.069 ± 0.010	−0.8	0.061 ± 0.004	0.041 ± 0.002	4.1
Exploration rate (gene)	0.049 ± 0.003	−5.6	0.031 ± 0.002	0.0226 ± 0.0009	4.6
Lifespan (gene)	3,645 ± 74	0.8	3,712 ± 48	3,815 ± 43	−1.6
Deepest lineage	530 ± 16	0.4	540 ± 16	533 ± 12	0.3
Situations met	131.8 ± 3.0	0.7	135.1 ± 3.6	135.2 ± 4.5	−0.0
Distinct behaviours ever seen	557 ± 21	5.2	751 ± 31	994 ± 45	−4.5
Behaviour diversity (bits)	4.36 ± 0.14	2.7	4.81 ± 0.09	4.53 ± 0.08	2.2

Nutrients do not move. Across a 2.25-fold range of world energy, the standing crop on land is 0.5350, 0.5334 and 0.5411 units a land cell. Taking 10,000 units out of `base` took 9,780 from detritus, 233 from bodies and nothing from the land. Adding 15,000 put 14,577 into detritus. The land is grazed down to the same density every time, and the rest of the world's energy is a reservoir sitting in the soil.

So population is more than proportionally sensitive to energy, not less. Over the measured range the relation is close to a straight line, population ≈ 0.0594 × energy − 614, which means about 10,300 units of overhead before the world supports a single agent and about 59 agents for each 1,000 units after that. Section 5.2's rule of thumb, "doubling `--energy` roughly doubles the population", is close at the top of this range and wrong at the bottom: 20,000 to 45,000 is 2.25 times the energy and 3.5 times the population.

This is the one place where a prediction written mid-run earned its keep. With `e20k` finished and `e45k` still running, the pool decomposition above was used to pre-register four numbers for the unfinished arm (`predictions.md`, 16:35). Nutrients, detritus and `mean_energy` all landed inside their stated bounds. The population interval, 1,600 to 2,000, did not: the answer was 2,066, although the straight line that generated the interval gave 2,002.

What the richer world does not buy. Lineage depth is flat across all three arms (530, 540, 533), so generations are not energy-limited; nor are situations met (132, 135, 135). What a richer world does buy is behaviours: the archive runs 557, 751, 994. And what it costs is learning, with both learning and exploration genes falling as the world gets richer. A denser population shares the same fixed standing crop, so each agent holds less (`mean_energy` 0.489, 0.451, 0.422) and lives a shorter, more crowded life in which learning pays less.

11.6 E6: does a bigger world sustain more variety?

No. It sustains more agents, at the same density, doing the same things.

`big` is 128 × 128 cells with 53,000 energy units, which is the same energy a cell as the default, and a 3,500-agent cap so that the cap cannot bind. At year 2,000 it holds 2,019 ± 13 agents against `base`'s 1,151 ± 9. The world has 1.79 times the land cells and 1.77 times the energy, and it holds 1.76 times

the population.

Table 13. The bigger world against the default, at year 2,000, mean of sixteen seeds (± standard error), with Welch's *t*.

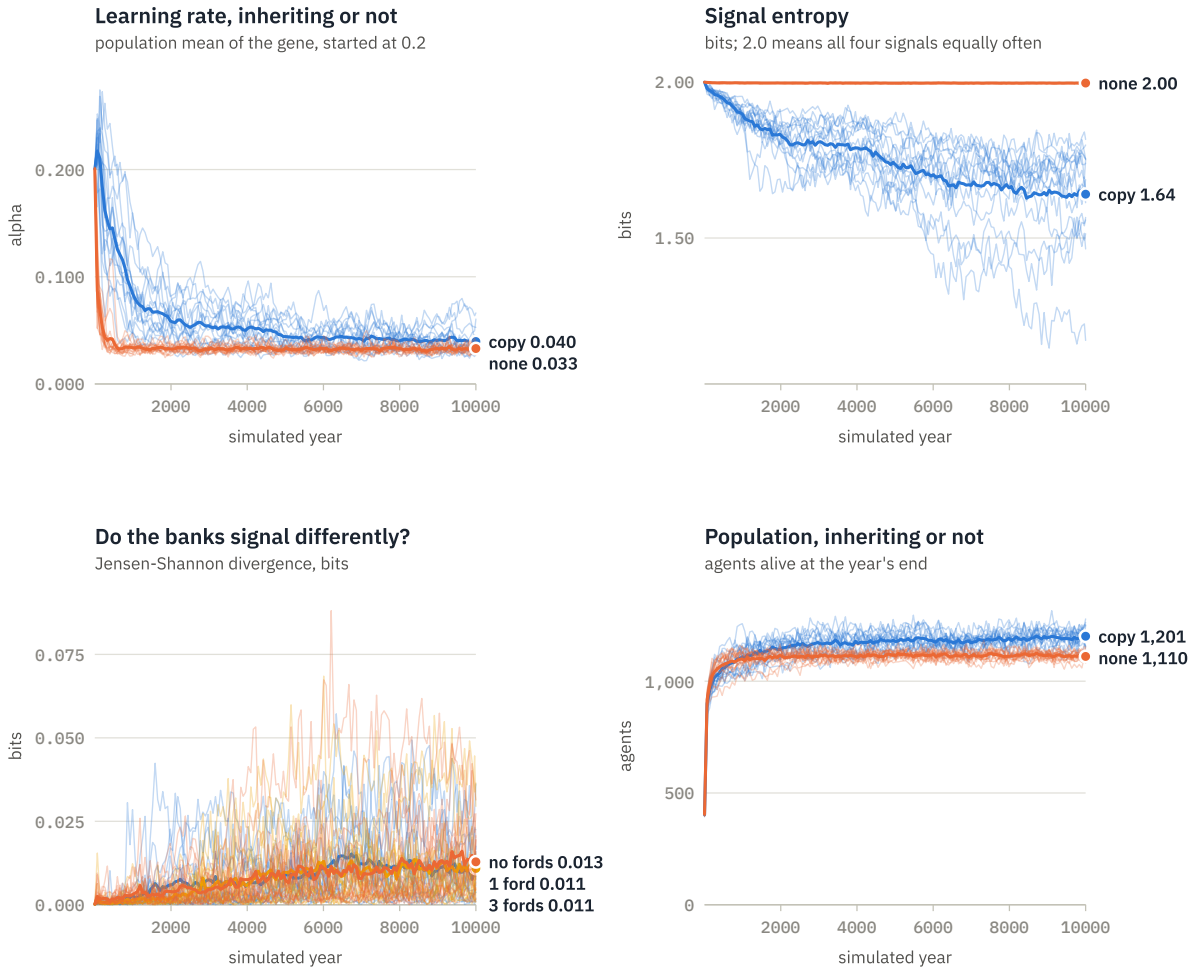
MEASURE	BASE (96 × 96)	BIG (128 × 128)	RATIO	*T*
Land cells	8,520	15,243	1.789	
Population	1,151 ± 9	2,019 ± 13	1.755	−54.1
Nutrients on land	4,544 ± 33	8,175 ± 69	1.799	−47.7
Standing crop a land cell	0.5334	0.5363	1.005	
Detritus	24,913 ± 40	43,870 ± 75	1.761	−223.1
`mean_energy`	0.451 ± 0.007	0.457 ± 0.005	1.013	−0.7
Distinct behaviours ever seen (of 2,048)	751 ± 31	805 ± 23	1.072	−1.4
Behaviour diversity (bits)	4.805 ± 0.094	4.840 ± 0.086	1.007	−0.3
Difference between the banks' signals	0.0060 ± 0.0018	0.0029 ± 0.0009	0.483	1.5
Signal entropy (bits)	1.8284 ± 0.0127	1.8298 ± 0.0120	1.001	−0.1
Policy complexity (of 20)	19.783 ± 0.038	19.770 ± 0.057	0.999	0.2
Deepest lineage	540 ± 16	613 ± 22	1.135	−2.6
Learning rate (gene)	0.061 ± 0.004	0.047 ± 0.004	0.777	2.4

The standing crop is invariant across size as well as across energy. 0.5334 units a land cell in the default world and 0.5363 in the big one, a difference of 0.5%. Section 11.5 found the same constant across a 2.25-fold range of energy. Two different ways of changing the world, and the land is grazed to the same density in all of them. That is the steadiest number in this section.

Nothing about variety improves. Behaviour diversity is identical to three significant figures. The archive rises 7%, which is inside seed noise at $t = -1.4$. The two banks' signals actually diverge *less* in the bigger world, not more as predicted, though at $t = 1.5$ that is not a result either. The only measures that move are population, which scales with the world, and lineage depth, which is 14% deeper.

And this is a fact about the world rather than about the measure, which is the one thing E6 was set up to distinguish. The archive sits at 805 of its 2,048 cells, so 1,243 descriptors have never been seen and the ceiling is nowhere near binding. A world nearly twice the size, running the same 2,000 years, simply does not produce behaviours it was not already producing. Section 7.2 worried that `novelty` would saturate because the descriptor space fills; the truth is less flattering to the world and more flattering to the measure. The space does not fill. The world stops exploring it.

11.7 Ten thousand years



Bold lines: the mean of sixteen seeds; faint lines: each seed. copy is the defaults; none starts every child with a blank Q-table; the river arms differ only in how many fords cross the river.

Figure 7. The four conditions carried on to year 10,000, sixteen seeds each. Top left: the learning-rate gap between inheriting and not inheriting closes to within seed noise. Top right: only the inheriting world ever structures its signalling. Bottom left: the river arms stay together for the whole run. Bottom right: inheriting what was learned supports 8% more agents, permanently.

Four of the nine conditions were carried on from 2,000 to 10,000 years, chosen by a trend test rather than by taste. Over years 1,500 to 2,000 `base` was still moving (signal entropy falling, $t = -3.2$; population rising, $t = 2.7$); `ford1` and `wall` were E3 and unfinished; `none` is the control without which the learning-rate question cannot be read. The other five were flat and stop at 2,000. Nine more predictions were written before the long runs started (Section 11.8, L1 to L9). The run took eight and a half hours overnight, 32 processes at a time, and every one of the 64 seeds finished.

Table 14. Copy against none at year 10,000: the mean over years 9,901 to 10,000, sixteen seeds a side, Welch's *t*. The year-2,000 values are from Section 11.2.

	COPY, YEAR 2,000	NONE, YEAR 2,000	COPY, YEAR 10,000	NONE, YEAR 10,000	*T* AT 10,000
`gene_alpha` (learning rate)	0.0607	0.0327	0.0391	0.0326	2.33
Ratio, copy over none	1.86		1.20		
`gene_eps` (exploration)	0.0310	0.0151	0.0212	0.0145	5.74
Signal entropy, bits	1.828	1.997	1.654	1.997	−8.90
Agents whose favourite signal is silence			83%	33%	
Population	1,151	1,105	1,198	1,111	9.81
Generations deep	540	382	2,462	1,793	14.40
Nutrients standing on the land	4,544	4,154	4,839	4,155	15.18
Crossings a year	232	319	183	318	−4.65

The Baldwin gap was a transient. The learning-rate ratio, inheriting over not, went 3.7 at year 300, 2.6 at 1,000, 1.86 at 2,000, 1.37 at 5,000 and 1.20 at 10,000, and *t* fell below 3 for the first time at year 9,000. Section 6 read the year-300 gap as evolution turning learning down when learning is not passed on. Over ten thousand years it looks the other way round: both worlds are heading to the same low learning rate, and the world that starts each child blank simply gets there in a few hundred years while the world that hands down its Q-tables takes thousands. The `none` arm reached 0.034 by year 1,000 and has sat at 0.032 to 0.034 ever since. Inheritance does not change where the learning rate ends up; it changes how long the journey takes. The exploration gene still separates at $t = 5.7$, so not every learning trait has converged, but it too has narrowed, from a ratio of 2.1 to 1.5.

What inheritance does do, permanently, is three things.

- **It is the only route to a signalling habit.** Without inheritance, signal entropy sits at 1.997 bits from year 1 to year 10,000: the four signals are used almost exactly equally, and silence is the favourite of 33% of agents against the 25% that chance would give. With inheritance, silence is the favourite of 83% by year 10,000. Speaking costs 0.05 energy a tick, a cost too small for one lifetime of Q-learning to detect against the noise of feeding, and large enough to show up once the estimate is carried down 2,400 generations. This is the one result in Section 11 that is cultural in the full sense: a behaviour no individual can learn, which a lineage learns. It is also why Section 11.3 found no dialects. The only thing inheritance teaches about the channel is to leave it quiet, and that lesson is the same on both banks.
- **It feeds more agents.** 1,198 against 1,111, 7.8% more, $t = 9.8$, and the gap was still there at every 1,000-year checkpoint from 2,000 on. The inheriting world also leaves more nutrients standing on the land and holds more energy in each body, so the extra agents come from grazing better, not from grazing harder, and the margin is still growing. The blank world's standing crop has sat at 4,153 to 4,155 units from year 2,000 to 10,000 and its mean body energy at 0.43; the inheriting world's crop has risen from 4,544 to 4,839 and its body energy from 0.451 to 0.483. Section 11.5 found the standing crop pinned at about 0.534 a land cell across three energy levels at year 2,000.

That holds across energy but not across time: an inheriting world keeps getting better at grazing for all ten thousand years, and a blank one is pinned from the start.

- **It turns generations over faster.** 2,462 generations against 1,793, so a generation lasts 4.1 years in the inheriting world and 5.6 in the blank one. Inheriting agents also wander across the river less, 183 crossings a year against 318.

The banks and the river. Signal entropy in `base` fell from 1.828 at 2,000 to 1.666 at 7,000 and then stopped: over the last 2,000 years its trend is −0.006 bits a millennium, $t = -0.6$. The drift into silence has found its floor at about 1.65 bits. Bank imbalance in `base` drifted down, from 0.268 at 2,000 to 0.173 at 10,000, and population crept up throughout, 1,151 to 1,198.

The river arms did not change their verdict. At year 10,000 `dialect_jsd` is 0.0098 with three fords, 0.0100 with one and 0.0130 with none, $t = 0.98$ for no fords against three. The walled world has been the highest of the three at each of the last three checkpoints, and at 9,000 and 10,000 all three arms line up in the order contact predicts for the first time. That is the direction isolation would predict, but the margin is about a third and inside seed noise; no seed in any arm is above 0.10 at the end, and the largest single seed-year in any arm in ten thousand years was 0.096, in the walled world at year 6,193. If drift is separating the walled banks, it is doing it at something like 0.001 bits a thousand years, and the paper's 0.2 would arrive somewhere beyond year 100,000. One river measure leans: the walled world's signal entropy is 1.752 against 1.654 ($t = -2.4$), below the $|t| = 4$ bar Section 11.9 sets for a result, but in the same direction at 2,000, 5,000 and 10,000. If it is real, contact speeds the drift into silence. The obvious mechanism is that a fully silent lineage on one bank can recolonise the other across a ford, and behind a wall it cannot.

Survival. No seed died out in any of the four arms in 10,000 years. The smallest population ever recorded was 318, in the founding years, and the smallest single bank 135; after year 2,000 no bank in any seed fell below 172. The world is robust over a horizon fifty times longer than Section 6 tested.

11.8 The predictions, scored

Twenty-seven predictions were written before the first world existed, plus four more written mid-programme when `e20k` had landed and `e45k` had not, and nine more before the four long runs were carried past year 2,000. None has been edited since. The full text sits in `_build/experiment-s7/predictions.md`, above a correction line that records what was already wrong before any result came in.

Table 15. Every pre-registered prediction and what happened. PARTLY means one clause of a compound prediction held and another did not.

#	PREDICTION	WHAT HAPPENED	
T1	Aggregate throughput 3 to 5 years/s	17.5, and 20.2 with two conditions at once	wrong
E1.1	Population within 10% across the three arms, all $\|t\| < 2$	within 4%, but $t = 4.5$ and 4.2	partly
E1.2	`q_states` ordered copy > half > none, all $\|t\| > 4$	135 / 136 / 66: `half` equals `copy`	wrong
E1.3	`policy_complexity` copy over none by at least 1.3 of 20	19.78 / 18.96 / 18.14, gap 1.64, ordered	right
E2.1	`gene_alpha` copy at or above 0.20, none at or below 0.055	0.061 and 0.033	wrong
E2.2	`gene_eps` copy above 0.035, none below 0.025, $\|t\| > 3$	0.031 and 0.015, $t = 9.2$	partly

#	PREDICTION	WHAT HAPPENED	
E2.3	`half` between the other two on both genes	not between on either, and outside both on `gene_eps`	wrong
E2.4	Evolution turns learning down when it is not passed on	holds at every year from 50 on, $t = 6.4$	right
E3.1	Three fords, 5,000 years: `dialect_jsd` below 0.005	0.0086, rising in all three arms alike	wrong
E3.2	`ford1`: 60 to 160 crossings a year, jsd above 3× base but below 0.05	62 crossings; jsd 0.0090, equal to `base` ($t = -0.15$)	partly
E3.3	`wall`: 0 crossings, jsd above 0.05, bank gap above 0.25 in half the seeds	0 crossings; jsd 0.0083, equal to `base`; 6 of 16	wrong
E3.4	No seed extinct; at most 2 of 16 `wall` seeds lose a bank	none extinct, no bank lost, smallest bank 141	right
E3.5	Section 7's own example, one ford above 0.2 by year 2,000, is wrong	0.0020 at year 2,000; never above 0.052 in any seed-year	right
E4.1	`calm` logs 0 events, `base` about 150 per 300 years	0 and 156	right
E4.2	`calm` diversity lower by at least 0.3 bits, archive by at least 50	diversity *higher* by 0.20, archive lower by 11	wrong
E4.3	`calm` `change` lower by at least 0.05	lower by 0.016	wrong
E4.4	`calm` population 2 to 8% higher	1.8% lower	wrong
E4.5	`novelty` does not separate the arms	0 in both	right
E5.1	`e20k` 0.72 to 0.85×, `e45k` 1.15 to 1.40× and below 1.5×	0.51× and 1.80×	wrong
E5.2	`gene_repro` 20.5 to 22.5 everywhere, all $\|t\| < 3$	21.28 to 21.48, but $t = 5.6$	partly
E5.3	`mean_energy` rises with energy, stays below 1.0	falls with energy; below 1.0	partly
E5.4	`gene_lifespan` does not order cleanly	$t = 0.8$ and -1.6	right
E6.1	`big` archive at least 15% up, diversity at least 0.2 bits up	7.2% and 0.035 bits, both inside noise	wrong
E6.2	`big` `dialect_jsd` at least twice `base`	half of `base`	wrong
E6.3	`big` population 1.4 to 1.9× `base`	1.755×	right
E6.4	A flat archive would indict the 2,048-cell ceiling, not the world	archive flat, ceiling nowhere near binding at 805	right
P2.1	`e45k` nutrients within 2% of 4,550, crop within 2% of 0.535	4,610 and 0.5411	right
P2.2	`e45k` detritus about 39,000	39,490	right
P2.3	`e45k` population 1,600 to 2,000	2,066	wrong
P2.4	`e45k` `mean_energy` below 0.451	0.4221	right
L1	The learning-rate gap closes: ratio 1.0 to 1.4 at year 10,000, $\|t\| < 3$	1.20, $t = 2.33$	right
L2	`none` flat: `gene_alpha` 0.028 to 0.040, signal entropy above 1.99	0.0326 and 1.9965	right
L3	`base` signal entropy falls to 1.65 to 1.78 by year 10,000	1.654, flat since 7,000	right

#	PREDICTION	WHAT HAPPENED	
L4	`base` bank imbalance stays 0.20 to 0.35	fell to 0.173	wrong
L5	`base` population 1,150 to 1,260, 2,400 to 3,000 generations	1,198 and 2,462	right
L6	`dialect_jsd` ordered base < ford1 < wall at every checkpoint from 2,000	ordered at 2 of nine checkpoints, the last two, inside noise	wrong
L7	`wall` `dialect_jsd` above 0.10 at year 10,000	0.0130	wrong
L8	No seed extinct in 10,000 years	none; smallest population 318	right
L9	64 processes within 5% of 20.15 years/s	15.56, 23% slower	wrong

The tally. Of the thirty-nine scored in Table 15, 17 were right, 5 partly right and 17 wrong. Split by when they were written: of the twenty-six scored rows written before any world existed, 9 were right, 5 partly and 12 wrong; of the four made mid-programme with one result in hand, 3 were right; of the nine made at year 2,000 with every short run analysed, 5 were right. The predictions got better as the evidence behind them grew, which is how it should go, and the ones that failed at every stage failed where the model's mechanism was not what the paper said it was: the river (E3, L6, L7), the disturbances (E4) and the energy budget (E5.1).

What the wrong ones were worth. The four E4 predictions failed together and in the same direction, and that pattern is what sent us to look at how big the disturbances actually are; a single wrong prediction would have looked like noise. E1.2 and E2.3 failed because `half` is not what its name says, and chasing that disagreement is what produced Table 10. E5.1 failed in the opposite direction from the reasoning behind it: the energy really does pile up in detritus as predicted, but the fixed standing crop makes population more than proportionally sensitive rather than less, and finding that out is what made the mid-run prediction P2 possible, three-quarters of which then held on an arm that had not yet finished.

Being wrong on paper costs nothing. Being wrong without having written it down first costs the result, because there is then no way to tell a finding from a story told afterwards.

11.9 What this section does not show

Everything here is about river world. Nine conditions on one simulation with one parameter set. The Baldwin-style result in 11.2 is a fact about these agents in this world, as Section 6 said of its own; nothing here generalises to learning and evolution in anything else.

Where you stop decides what you find. That is the sharpest lesson of 11.2 and 11.7. The learning-rate ratio peaks at 3.7 near year 300, is 1.7 by year 2,000 and 1.2 by year 10,000. Stopped at 300 it is a large permanent effect; stopped at 10,000 it is a difference in speed. Even 10,000 years is one horizon: the inheriting world's standing crop and body energy were still rising when it stopped. A measurement at one horizon is a measurement at one horizon.

Sixteen seeds and twenty-two columns means some of these *t* values are chance. Each comparison in this section tests about twenty-two columns at once. With $|t|$ above 2 as the bar, roughly one column a comparison will clear it by luck alone. Everything claimed here as a result sits above $|t| = 4$, and most

above 6; the numbers quoted near 2, such as the archive in Table 9 or behaviour diversity in Table 12, are noted but not leaned on.

The measures are cruder than the numbers look. `novelty` reaches exactly 0 by year 300 in every arm and stays there, with the archive saturated around 750 of its 2,048 cells, so it discriminates nothing after the first few centuries. `q_states` counts a visited flag and, as Table 10 shows, that flag can outlive the knowledge behind it entirely. `ecology_entropy` and `archive_size` both inherit the same six-part descriptor, and Section 7.2's warning about it stands: these are the observer, and a better observer would see a different world.

Two arms are not quite the same program. `ford1` and `wall` each differ from the shipped kit by one line. The terrain was checked, and nothing else in those folders differs, but they are copies and should be read as such.

E4 is unanswered rather than answered. The null in 11.4 is a fact about disturbances that touch a 0.1% of the energy budget. It says nothing about whether disturbance would matter if it were large enough to notice.

And the most useful warning of the lot: check what a treatment does in the code before building an experiment on its name. `--inherit half` sounds like a halfway point between the other two arms, and E1 added it on that basis. It is not one. One line of reading would have caught it before 31 minutes of compute, and the only reason it became a result rather than a wasted arm is that the three arms disagreed in a pattern strange enough to send us back to the source.

Appendix A: Parameters

All of these are constants or options in `river_world.py`. Options can be set on the command line; constants need an edit (in a copy, Section 7).

Table A1. River world's parameters and their defaults.

PARAMETER	DEFAULT	WHERE
World size	96 × 96 cells, wrapping north to south, walled east and west	`--size`
Total energy	30,000 units, conserved	`--energy`
Agent capacity	2,000 at once	`--capacity`
Starting population	300 agents, 40 energy each, on random land cells	`World.__init__`
Terrain	rock 5% of cells, clearings 22%, forest the rest; the river is carved through them	`World.__init__`
River	3 cells wide; centre swings up to about 11 cells either side; fords at rows 16, 48 and 80, three rows deep	`World.__init__`
Time	day 20 ticks (night is the last 8); season 100 ticks; year 400 ticks	`DAY`, `SEASON`, `YEAR`
Costs	rest 0.5, move 0.9, ×1.25 at night, +0.05 to send a signal	`World.step`
Eating	up to 4 a tick; at most 150 held	`FEED`, `MAX_E`
Decomposition	5% (forest), 3.5% (clearing), 2% (ford) of detritus a tick; ×1.2 spring, ×1.0 summer, ×0.8 autumn, ×0.4 winter	`World.step_environment`
Diffusion	2% of a cell's nutrients a tick to each land neighbour	`World.step_environment`
River	erosion 0.2% of bank nutrients a tick; 20% of each row's load moves one row south a tick; 1% of the load settles as silt a tick, ×2.5 in spring	`World.step_environment`
Flood	25% chance each spring; 40 ticks of 6× silt and 3× erosion	`World.maybe_event`
Drought	15% chance each summer; 100 ticks of the river at a quarter of its flow and silt	`World.maybe_event`
Fire	12% chance at the start of summer and of autumn; radius 6; nutrients become detritus; agents in it lose half their energy	`World.maybe_event`
Learning	tabular Q-learning, 240 states × 20 actions, discount 0.95, reward = change in energy, random tie-breaks	`World.step`
Starting genes	lifespan 1,200 ticks, reproduce above 70 energy, mutation scale 0.05, learning rate 0.2, exploration 0.1	`World.__init__`
Mutation	each gene moves by a normal step × (mutation scale ÷ 0.05) × 80 ticks, 8 energy, 0.01, 0.05 and 0.03 respectively, within fixed limits	`G_SCALE`, `G_MIN`, `G_MAX`
Inheritance	the parent's Q-table is copied to the child	`--inherit copy`, `half` or `none`
Checkpoint	every 10 years, and on exit	`--save-every`

Appendix B: How the kit was tested

Everything below was run on the laptop before this paper was written, and every test passed in its final form.

- **Energy.** Conservation is asserted after every tick in every run. In the eight 300-year runs of Section 6 the largest end-of-year drift was 3.1×10^{-8} energy units, about one part in 10^{12} of the total.
- **Resume.** A run stopped and resumed produced a log byte-identical to an uninterrupted run in four cases: stopped at a year boundary; stopped part-way through a year; stopped with Ctrl-C at year 8 of 20; and killed outright with `kill -9` after its last checkpoint, where the resumed run trimmed the one log row written after the checkpoint and carried on.
- **Symmetry.** A world and its exact mirror image, left for right, were run side by side. The first version of the code favoured the east bank, by between three and seven to one, because untried actions were chosen in index order and east comes before west. Random tie-breaking removed the bias: which bank leads now varies with the seed and the mirror, as it should.
- **Review.** A line-by-line review found seven defects in the first version, all fixed and re-tested: an interrupted run skipped the seasonal event roll and could not resume exactly; the signal colouring of maps was always blank; maps of extinct worlds crashed; one failed seed stopped the summary; extinct seeds were hidden in the summary; a log row still being written crashed it; and a hard kill could leave duplicate rows in the log.
- **Tools.** `run_seeds.sh`, `summarise.py` and `snapshot.py` were run on real and on deliberately extinct worlds.

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